llmXive Automated Review of KronQ: LLM Quantization via Kronecker-Factored Hessian

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a robust technical argument for incorporating gradient covariance into the quantization objective, with sound mathematical derivations and consistent internal logic. However, there are minor discrepancies between the summary claims in the abstract and main text versus the specific numerical results reported in the primary tables.

First, the abstract and Section 5.1 state that KronQ achieves a perplexity of 7.93 on LLaMA-3-70B at 2-bit quantization. This value is found in the Appendix (Table 1 in `weight_only_zs`), but the primary results table (Table 1 in the main text) reports 8.43 for the same configuration. Since the main table is the primary evidence, the abstract and text should align with it or explicitly clarify the source of the discrepancy.

Second, the abstract claims that GPTQ and GPTAQ produce "degenerate quantizations (>2000 perplexity)" on LLaMA-3-70B. The data shows GPTQ at 2560.14 (consistent with the claim) but GPTAQ as "NaN" (indicating divergence). Conflating a specific high perplexity value with a complete divergence in the abstract slightly misrepresents the distinct failure modes observed.

Finally, the claim that KronQ "nearly doubles" the GPTAQ score on LiveCodeBench is an understatement; the data shows improvements of over 2x for both models tested. Using "more than doubles" would be a more precise description of the results.

These issues are minor and fixable through text edits, but they affect the precision of the paper's claims relative to its evidence.

Action items.

- [**writing**] Abstract claims GPTQ/GPTAQ produce '>2000 perplexity' on LLaMA-3-70B at 2-bit, but Table 1 shows GPTQ at 2560.14 and GPTAQ as 'NaN'. Clarify that GPTQ yields ~2560 PPL while GPTAQ diverges (NaN) to accurately reflect the distinct failure modes.
- [**writing**] Abstract and Section 5.1 cite 7.93 perplexity for LLaMA-3-70B at 2-bit, but the primary Table 1 reports 8.43. The 7.93 value appears only in the Appendix. Update the abstract and text to cite the primary table value (8.43) or explicitly distinguish the sources.
- [**writing**] Section 5.1 states KronQ 'nearly doubles' GPTAQ on LiveCodeBench, but Table 4 shows >2x gains (37.3 vs 16.3 and 22.3 vs 9.6). Change 'nearly doubles' to 'more than doubles' to accurately reflect the magnitude of improvement shown in the data.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure is visually clear and effectively communicates the data, but the caption contains a grammatical error where the third method (KronQ) is omitted from the text list, leaving the sentence incomplete.

Figure 2

The figure effectively visualizes the incoherence and weight distribution changes, but the caption contains a typo regarding the incoherence symbol, and the matrix plots in panel (b) use an inverted y-axis that may confuse readers.

Figure 3

The figure effectively visualizes the sensitivity rankings described in the caption, but the y-axis labels use ambiguous single-letter abbreviations without a legend, and some labels suffer from a rendering artifact that adds a vertical bar.

Figure 4

The figure effectively compares perplexity across methods, but the non-uniform x-axis scale is misleading, and the inset plot lacks a complete x-axis scale for the zoomed-in data points.

Figure 5

The figure presents absolute values on the axes while the caption claims the data is relative to a baseline, creating a contradiction. Additionally, the W2 data series defined in the legend is not visible in the chart.

Figure 6

The figure presents a perplexity vs. bit-width comparison but lacks a descriptive caption to identify the model or methods. Additionally, the inset plot's x-axis is poorly defined, and the main axis labels are cluttered.

Figure 7

The figure content appears to be a duplicate of Figure 2(b) (showing LLaMA-2-7B weight distributions) rather than the LLaMA-2-70B data described in the caption. The visual data (8192 channels) contradicts the caption's claim of 70B model data.

Action items.

- [**writing**] Figure 1 caption: The phrase 'GPTQ, GPTAQ, and on LLaMA-2-13B' is grammatically incomplete; it omits the third method name before the model list, though the legend identifies KronQ.
- [**writing**] Figure 1 caption: The list of methods 'GPTQ, GPTAQ, and KronQ' is grammati-

cally incomplete; it omits the third method name before the model list, though the legend identifies KronQ.

- [**writing**] Figure 2(a) caption contains a typo '$$-incoherence' instead of the symbol 'μ-incoherence' shown on the plot axis.
- [**science**] Figure 2(b) y-axis labels (0, 1024, 2048, 3072, 4096) are inverted (0 at top, 4096 at bottom), which contradicts the standard convention for matrix visualization where row 0 is typically at the top.
- [**writing**] Figure 3: The y-axis labels use single letters (o, k, q, v) which are ambiguous without a legend or explicit definition in the caption stating they correspond to the output, key, query, and value projections.
- [**writing**] Figure 3: The labels '#4 o', '#5 k', etc., are rendered with a vertical bar artifact (e.g., 'k|', 'q|') likely due to a font rendering error, making them look like 'k|' or 'q|' instead of just the letter.
- [**science**] Figure 4: The x-axis labels (2.0, 2.17, 2.29, 2.43, 2.57, 3.0) are non-uniform and do not correspond to the visual spacing of the data points, making the 'Average Bit-width' scale misleading and difficult to interpret.
- [**writing**] Figure 4: The inset plot's x-axis is labeled only with '3.0', failing to show the full range of bit-widths for the zoomed-in region, which limits the utility of the inset for comparing the methods at that specific scale.
- [**science**] Figure 5: The caption states the data is 'relative to the bf16 baseline,' but the y-axes are labeled with absolute units ('Peak VRAM (GB)' and 'TPOT (ms/token)') and the bars show absolute values (e.g., ~100 GB) rather than relative ratios. This contradicts the caption's description of the data presentation.
- [**writing**] Figure 5: The legend in panel (a) includes 'W2', but the corresponding dark blue bars are not visible in the chart, likely obscured by the 'W4' bars or missing entirely, making the data for W2 unreadable.
- [**writing**] Figure 6: The caption is explicitly '(no caption)', leaving the plot unexplained and failing to define the context (e.g., model size) or the specific methods being compared.
- [**science**] Figure 6: The inset plot's x-axis is labeled '3.0' without a range or tick marks, making it impossible to determine the bit-width scale or the specific data points being highlighted.
- [**writing**] Figure 6: The x-axis label 'Average Bit-width' is present, but the axis ticks (2.0, 2.17, etc.) are rotated and crowded, reducing readability.
- [**writing**] Figure 7: The provided image is a 3-panel plot showing 'Original', 'H_X only', and '$H_X + H_G$' weight distributions with $CV_{in/out}$ metrics, which matches the content of Figure 2(b) in the paper, not Figure 7. The caption 'Weight distributions of LLaMA-2-70B' does not match the visual content shown.

- [**science**] Figure 7: The figure displays data for LLaMA-2-7B (inferred from the 8192 channel dimension and similarity to Figure 2), but the caption claims it shows LLaMA-2-70B. This is a factual mismatch between the visual data and the caption.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript demonstrates excellent accessibility for a competent reader from an adjacent field (e.g., optimization, numerical linear algebra, or general deep learning). The authors consistently define all non-standard acronyms, symbols, and method names at their first occurrence.

Specifically, the paper handles its specialized vocabulary well:

- **Acronyms:** Terms like "PTQ" (post-training quantization), "OBS" (Optimal Brain Surgeon), "K-FAC" (Kronecker-factored Approximate Curvature), and "BiIP" (Bidirectional Incoherence Processing) are explicitly expanded upon first use. Even newer or more specific terms like "LDLQ" are introduced with sufficient context or referenced to prior work where the definition is standard.
- **Notation:** The mathematical notation is rigorous and self-contained. Variables such as $\mathbf{H}_X$ (input activation covariance) and $\mathbf{H}_G$ (gradient covariance) are defined immediately before or within the equations where they appear (e.g., Section 3.1, Eq. 1 and Eq. 2). The dimensions of matrices (e.g., $\mathbf{X} \in \mathbb{R}^{d_{\text{in}} \times n}$) are clearly stated in the Preliminary section.
- **Method Names:** References to specific algorithms (GPTQ, GPTAQ, QuIP, QuIP#, etc.) are accompanied by brief, one-sentence descriptions of their function or the specific limitation they address, ensuring a reader unfamiliar with the specific subfield of LLM quantization can follow the comparative logic.
- **Buzzwords:** Terms like "incoherence" are not used as vague buzzwords; they are given a precise mathematical definition (μ) in Section 3.2 (Eq. 4) and explained intuitively.

There are no instances of undefined symbols, unexplained acronyms, or "lab slang" that would stall a reader with a strong PhD in a neighboring discipline. The paper successfully bridges the gap between the specific subfield of LLM quantization and the broader machine learning community.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper's logical structure is sound and internally consistent. The central argument—that incorporating the gradient covariance $\mathbf{H}_G$ via the Kronecker-factored Hessian approximation improves quantization—follows logically from the premises. The derivation of the quantization objective (Eq. 6) correctly applies the K-FAC assumption, and the subsequent claim that $\mathbf{H}_G$ cancels in the column-wise OBS update (Proposition 1) is mathematically entailed by the provided proof in the Appendix.

The experimental claims are consistent with the presented data. The abstract's claim that GPTQ/GPTAQ diverge on LLaMA-3-70B at 2-bit while KronQ succeeds is directly supported by Table 3 (showing NaN/2560.14 vs 8.43) and the text in Section 5.1. The mixed-precision argument (Section 5.3) correctly links the proposed sensitivity metric $s_\ell = \mathrm{tr}(\mathbf{H}_G) \cdot \mathrm{tr}(\mathbf{H}_X)$ to the observed improvement in perplexity over activation-only baselines, as evidenced by Table 4 and Figure 4.

Definitions and notation remain stable throughout: $\mathbf{H}_X$ and $\mathbf{H}_G$ are consistently defined as input activation and gradient covariances, respectively. The scope of claims is appropriately bounded; for instance, the "generalization" claims in the abstract and conclusion are qualified by the specific models and bit-widths tested (LLaMA-2/3, W2/W3/W4), matching the experimental setup in Section 5.1. No contradictions were found between the limitations section and the results, nor between the method description and the algorithm pseudocode. The logical flow from problem identification (input-only Hessian) to solution (KronQ) to validation (experiments) is coherent.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper presents a compelling method for incorporating gradient covariance into post-training quantization, with strong empirical results on the LLaMA family. However, there are minor instances of overreach in the framing of the results' scope, particularly regarding the universality of the "divergence" problem in baselines and the breadth of generalization claims.

In the **Abstract**, the statement that GPTQ and GPTAQ "diverge or produce degenerate quantizations" on LLaMA-3-70B is factually correct based on the provided tables. However, the phrasing implies this is a general property of these methods at 2-bit, whereas the results in Table 1 show that GPTQ and GPTAQ function correctly (albeit with higher perplexity) on LLaMA-2-7B and LLaMA-2-13B at 2-bit. The divergence is a specific artifact of the LLaMA-3-70B weight distribution (as explained in Appendix A.6). The abstract should be slightly narrowed to specify that this failure mode is observed on LLaMA-3-70B, preventing the reader from inferring that 2-bit quantization is universally impossible for GPTQ/GPTAQ on all architectures.

In the **Introduction**, the claim of "consistent state-of-the-art results" is largely supported, but the magnitude of the "state-of-the-art" status varies significantly. On LLaMA-2-70B at W4, the improvement over GPTQ is marginal (3.41 vs 3.50), whereas on LLaMA-3-70B at W2, it is transformative (8.43 vs divergence). The current phrasing risks conflating these two regimes. While not a fatal flaw, clarifying that the "consistent" gains are most pronounced in the ultra-low-bit regime where baselines struggle would better align the rhetoric with the nuanced evidence.

Finally, the **Generalization** claims in Section 5 and the **Conclusion** extend the findings to "newer model families" and "harder benchmarks." While the paper does test Gemma-3, DeepSeek, and Phi-4, the evaluation is restricted to specific bit-widths (W4/W2) and a limited set of benchmarks (MMLU, GPQA, AIME, LiveCodeBench) on only two of these new models. The rhetoric suggests a broader validation than the specific experimental setup provides. Adding qualifiers such as "on the tested newer model families" or "across the evaluated reasoning benchmarks" would ensure the scope of the claim matches the specific evidence presented.

These are primarily issues of precision in the framing of the results rather than fundamental flaws in the evidence. The paper's core contribution is well-supported, but tightening the scope of the general claims will prevent overinterpretation by the reader.

Action items.

- [**writing**] Abstract: The claim that GPTQ/GPTAQ 'diverge' at 2-bit is specific to LLaMA-3-70B (Table 3), yet the abstract implies a general failure of these methods. Table 1 shows they work on LLaMA-2 models. Scope the claim to 'on LLaMA-3-70B' to avoid implying universal divergence.
- [**writing**] Introduction: The claim of 'consistent state-of-the-art results' overstates the marginal gains on LLaMA-2-70B (W4). Clarify that gains are 'particularly significant at ultra-low bit-widths where baselines fail' to match the evidence in Tables 1 and 3.
- [**writing**] Section 5 & Conclusion: Claims of generalization to 'newer model families' and 'harder benchmarks' rely on limited tests (2 models, specific bits). Add qualifiers like 'on the tested newer model families' to align the scope with the specific evidence provided.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper presents a post-training quantization (PTQ) method for Large Language Models (LLMs) that incorporates gradient covariance into the quantization objective. The work is purely algorithmic and computational in nature, focusing on model compression and efficiency.

A review of the manuscript for safety and ethics risks yields no actionable concerns:

1. **Data Provenance:** The method uses standard, public calibration datasets (WikiText-2) and publicly available model weights (LLaMA-2, LLaMA-3, etc.). There is no use of private,

sensitive, or scraped personal data requiring consent or IRB approval.

2. **Dual-Use Risk:** While the resulting quantized models are more efficient and could theoretically be deployed in various contexts, the method itself (KronQ) does not lower the barrier to a specific harmful capability (e.g., generating disinformation, bypassing safety filters, or creating biological agents) in a way that differs from standard quantization techniques. The paper does not describe a system designed to deceive, surveil, or manipulate.
3. **Vulnerability Disclosure:** The paper does not report a security vulnerability in a live system or an operational exploit; it reports a performance improvement in a compression algorithm.
4. **Bias and Fairness:** The evaluation focuses on perplexity and standard commonsense reasoning benchmarks. The paper does not claim to address or introduce specific demographic biases, nor does it present results that suggest a new, unmitigated fairness harm to an identifiable group.

The research falls squarely within the domain of low-risk systems optimization. No specific disclosures, mitigations, or ethics statements are required beyond standard academic practice for this type of work.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling method for incorporating gradient covariance into post-training quantization, but the experimental evidence relies heavily on single-run results without reported variance. Table 1 and Table 3 present headline perplexity and accuracy numbers (e.g., 8.15 vs 9.81 on LLaMA-2-7B at W2) as definitive facts. In the context of post-training quantization, where performance can fluctuate based on random seeds, calibration data ordering, or floating-point non-determinism, a single run is insufficient to establish that the observed gains are robust. The absence of standard deviations or confidence intervals makes it impossible to distinguish a genuine methodological improvement from statistical noise or a lucky initialization.

Furthermore, the dramatic claim that GPTQ and GPTAQ “diverge” or produce “degenerate” quantizations at 2-bit on LLaMA-3-70B, while KronQ succeeds, is based on a single execution. Divergence in these algorithms can sometimes be sensitive to specific random seeds or the order of layer processing. To support the claim that KronQ is fundamentally more stable, the authors should demonstrate that the baselines consistently fail and KronQ consistently succeeds across multiple random seeds.

Finally, the ablation study in Table 4 effectively isolates the BiIP components but conflates the base quantizer (GPTQ vs. GPTAQ) with the proposed method. The table compares “KronQ

(GPTAQ base)" against "GPTQ base" and "GPTQ base + BiIP". To rigorously isolate the contribution of the GPTAQ drift correction mechanism itself, a control run is needed: "KronQ with GPTQ base + BiIP" versus "KronQ with GPTAQ base + BiIP". Without this, the specific contribution of the asymmetric drift correction to the final gain remains partially confounded with the BiIP improvements.

Action items.

- **[writing]** The paper presents a compelling method for incorporating gradient covariance into post-training quantization, but the experimental evidence relies heavily on single-run results without reported variance. Table 1 and Table 3 present headline perplexity and accuracy numbers (e.g., 8.15 vs 9.81 on LLaMA-2-7B at W2) as definitive facts. In the context of post-training quantization, where performance can fluctuate based on random seeds, calibration data ordering, or floating-point non-determinism, a

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The paper presents a novel quantization method (KronQ) and reports extensive empirical results across multiple models and bit-widths. However, the statistical reporting of these results is insufficient for rigorous evaluation.

Missing Uncertainty Reporting: Throughout the results sections (Sections 4.1, 4.2, and Appendices), all quantitative metrics (WikiText-2 perplexity, zero-shot accuracy) are reported as single point estimates (e.g., "5.56", "79.2%"). There is no mention of the number of random seeds used for calibration, nor are standard deviations (SD), standard errors (SE), or confidence intervals (CI) provided. In deep learning quantization, results can exhibit variance due to the stochastic nature of the calibration dataset sampling or numerical instabilities in the optimization process. Without reporting variance (e.g., "5.56 ± 0.02"), it is impossible to determine if the reported improvements over baselines are statistically significant or merely due to random fluctuation. The field standard for such comparisons is to report mean ± SD over at least 3 independent runs.

Lack of Statistical Significance Testing: The paper makes strong comparative claims, such as KronQ achieving "strictly better" perplexity or baselines "diverging" (PPL > 2000). While the magnitude of the difference in the "divergence" cases is large enough to be practically significant, the smaller but consistent gains in other settings (e.g., 5.56 vs 5.69) are presented without any statistical test (e.g., paired t-test) or confidence intervals to support the claim of superiority. The absence of uncertainty metrics prevents the reader from assessing the reliability of these marginal improvements.

Recommendation: The authors should re-run their experiments with multiple random seeds (e.g., 3-5) for the calibration set and report the mean and standard deviation for all perplexity and accuracy metrics in the main tables. If the results are deterministic given the fixed seed, this should be explicitly stated, and the potential for seed-dependent variance should be acknowledged as a limitation. Adding error bars or a "± SD" column to the tables is a minimal change that would significantly strengthen the statistical validity of the claims.

Action items.

- [**writing**] Table 1 and Table 2 report single-point perplexity and accuracy values (e.g., '5.56', '79.2') without any measure of uncertainty (standard deviation, standard error, or range). Given that quantization results can vary with calibration seeds or numerical noise, report mean ± SD over at least 3 independent runs for all reported metrics to establish stability.
- [**writing**] The abstract and Section 4.1 claim GPTQ/GPTAQ 'diverge' or produce 'degenerate' results (e.g., PPL > 2000) while KronQ achieves low PPL. While the magnitude difference is clear, the paper lacks a formal statistical test or confidence intervals to confirm that the observed gains (e.g., 7.93 vs >2000) are not artifacts of a single lucky/unlucky calibration seed. Report variance across seeds or explicitly state that results are deterministic under the fixed seed used.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates high writing quality, allowing a reader to move through the technical argument with minimal friction. The abstract effectively summarizes the problem, the proposed method (KronQ), and the headline results, including the specific perplexity gains on LLaMA-3-70B. The introduction clearly establishes the limitation of existing methods (ignoring $\mathbf{H}_G$) and outlines the contributions with precise topic sentences.

Throughout the paper, the logical flow is maintained. The transition from the Preliminary section to the Method section is smooth, with the Kronecker-factored approximation introduced naturally as a solution to the stated problem. Paragraphs are well-structured, typically opening with a clear statement of the point to be made (e.g., the motivation for bidirectional incoherence processing) followed by the necessary mathematical derivation or empirical evidence.

The prose is concise and professional. Technical terms are used consistently (e.g., $\mathbf{H}_X$, $\mathbf{H}_G$, BiIP), and the distinction between the base quantizer (GPTAQ) and the proposed enhancements is maintained without confusion. Transitions between sections, such as moving from the theoretical derivation of the sensitivity metric to the experimental validation of mixed-precision allocation, are handled effectively with signposting sentences.

While the paper is dense with mathematical notation, the surrounding text successfully guides the reader through the derivations, explaining the intuition behind equations like the Kronecker-factored objective and the cancellation of $\mathbf{H}_G$ in the update rule. The experimental section is well-organized, with clear subheadings for different quantization regimes and ablation studies that directly address the components of the proposed method. There are no instances of garden-path sentences, ambiguous pronoun references, or structural ordering issues that would force a reader to re-read a passage to grasp the meaning. The writing is clear, direct, and effectively communicates the paper's contributions.